1. Item 10 (Most relevant)

> **Make sense of, relate and use a variety of representations when interacting with a problem**

How it is reflected:

- The problem requires students to understand and use multiple forms of representation (e.g. textual descriptions combined with tables, graphs, or numerical data).
- These representations must be meaningfully connected to the problem goal rather than interpreted in isolation.

👉 This is the **core competency** assessed by the question.

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2. Item 11

> **Switch between different representations in the process of finding solutions**

How it is reflected:

- Students must move back and forth between the real-world context (wind speed, power output, efficiency, etc.) and mathematical representations (data, numerical relationships).
- For example: reading values from a graph → translating them into quantitative judgments → returning to the real-world decision context.

👉 Representation switching is a **key step** in the solution process.

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3. Item 13

> **Use an understanding of the context to guide or expedite the mathematical solving process**

How it is reflected:

- Students must understand real-world constraints of wind power generation (e.g. variability of wind speed, feasibility, realistic ranges).
- Mathematical judgments must be contextually reasonable, not just formally correct.

👉 Contextual understanding directly affects the **validity of the final answer**.

4. Item 12

> **Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation**

How it is reflected:

- The solution typically involves:
  1. Reading and interpreting information
  2. Comparing or estimating values
  3. Integrating results
  4. Drawing a conclusion

👉 The problem requires **structured, multi-step reasoning**, not a single operation.

5. Item 7

> **Manipulate numbers, graphical and statistical data and information**

How it is reflected:

- Students must extract numerical information from tables or graphs;
- They must compare values or identify trends in the data.

👉 While the calculations are not complex, **data handling** is essential.

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6. Item 5

> **Understand and utilise constructs based on definitions, rules and formal systems**

How it is reflected:

- Correct interpretation of quantities used in the problem (e.g. power, efficiency, rate of change);
- Interpreting data according to accepted mathematical and scientific conventions.

👉 This reflects a sound understanding of **formal quantitative concepts**.

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7. Item 2

> **Select an appropriate strategy from a list**

How it is reflected:

- Students must decide whether to solve the problem through comparison, estimation, or direct information extraction;
- Different strategies may influence efficiency.

👉 Strategic choice is present, but **not the main challenge**.

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8. Item 1

> **Perform a simple calculation**

How it is reflected:

- The problem may involve simple arithmetic operations such as addition, subtraction, or comparison;
- The computational demand is low.

👉 This is a **supporting skill**, not the focus of assessment.

① 第 10 项（最相关）

> **Make sense of, relate and use a variety of representations when interacting with a problem**

具体体现：

- 题目要求学生理解并使用不同形式的信息表示（如文字说明 + 表格 / 图像 / 数据）。
- 学生需要将这些表示与问题目标联系起来，而不是孤立地读取数据。

👉 这是该题的**核心能力要求**：理解并整合多种信息表示。

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② 第 11 项

> **Switch between different representations in the process of finding solutions**

具体体现：

- 学生需要在现实情境（风速、发电量、效率等）与数学表示（数据、数值关系）之间来回切换。
- 例如：从图表中读取信息 → 转化为数量判断 → 回到现实问题做决策。

👉 表征之间的"转换"是解题关键步骤之一。

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③ 第 13 项

> **Use an understanding of the context to guide or expedite the mathematical solving process**

具体体现：

- 学生必须理解风力发电的现实约束（如风速变化、可行性、合理范围）。
- 数学判断需要符合实际情境，而非只追求形式上的计算结果。

👉 情境理解直接影响答案是否合理。

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④ 第 12 项

> **Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation**

具体体现：

- 解题过程通常包括：
  1. 读取信息
  2. 比较或计算
  3. 综合判断
  4. 得出结论

👉 并非一步完成，而是**结构清晰的多步骤推理**。

⑤ 第 7 项

> **Manipulate numbers, graphical and statistical data and information**

具体体现：

- 学生需要从表格或图像中提取数值信息；
- 对数据进行比较、判断大小或趋势。

👉 虽然计算不复杂，但对数据的处理是必要的。

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⑥ 第 5 项

> **Understand and utilise constructs based on definitions, rules and formal systems**

具体体现：

- 正确理解题目中涉及的数量含义（如功率、效率、变化关系）。
- 按数学和科学中约定的方式解读数据。

👉 体现对数学与情境中“量”的规范理解。

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⑦ 第 2 项

> **Select an appropriate strategy from a list**

具体体现：

- 学生需要判断是通过比较、估算还是直接读取信息来解决问题。
- 不同策略会影响解题效率。

👉 有策略选择，但不是主要难点。

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⑧ 第 1 项

> **Perform a simple calculation**

具体体现：

- 可能涉及简单的加减、比较或近似判断。
- 计算本身难度较低。

👉 属于支持性能力，而非核心考查点。